Physics 111 - Experiment 1: The Large Amplitude Pendulum
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We are trying to measure the amplitude dependence of the period of a simple pendulum.
By extrapolating to zero-amplitude we determine g.

Procedure
We measured the period of a simple pendulum of known length as a function of release angle.
1. Rather than measure the pendulum length directly, we used a meter stick with caliper jaws to measure from the support to the bottom of the pendulum ball. We measured the diameter of the ball with calipers, and found the distance from the support to the center of the ball by subtraction.
2. To get the angle we measured the distances s and d and used the relation sin(theta) = d/s. This seemed more accurate than using the small protractor provided.
3. The actual period was measured for the first swing after release, using a photogate timer in pendulum mode placed near the bottom of the trajectory.

Data
total length = 0.757 m +/- 0.002 m
ball diameter = 0.0256 m +/- 0.0005 m
pendulum length = 0.744 m +/- 0.002 m
string length s = 0.731 m +/- 0.002 m

The period was measured three times for each angle and averaged:

theta (rad)	d (m)	T (s)	aver T (s)
0.104 +/- 0.002	0.076	1.7326, 1.7330, 1.7328	1.7328
0.143	0.106	1.7294, 1.7296, 1.7297	1.7296
0.043	0.031	1.7286, 1.7288, 1.7288	1.7287
0.226	0.169	1.7380, 1.7381, 1.7382	1.7381
0.385	0.300	1.7606, 1.7609, 1.7609	1.7608
0.555	0.462	1.7460, 1.7460, 1.7462	1.7461

Analysis
We used Graphical Analysis to plot T vs theta^2 and fit a straight line. The straight line is an acceptable fit, showing that the relationship T = T0 * (1 + theta^2/16) is reasonable.
To get g we use the results from the program:
intercept = T0 = 1.728 +/- 0.0002 s
slope = T0/16 = 0.108 +/- 0.001 s => T0 = 1.73 +/- 0.02 s
and the pendulum equation T0 = 2*pi*sqrt(l/g) to find g = 9.84 m/s^2. The length of the pendulum is the largest uncertainty, at 0.3%, so it determines the uncertainty in
g = 9.84 +/- 0.03 m/s^2.

Comments
Our results are consistent with the claims in the lab manual. The value of g is about the same as we found in the free-fall experiment.